

School of Engineering

Cochin University of Science & Technology

B.Tech Degree 2 nd semester First Internal Examination, April 2024

23-200-0202 A Engineering Physics (2023 scheme)

(CE A, CE B/ME B, SF)

Time: 2 hours

Max.Marks: 30

Course Outcomes:

1. Interpret modern devices and technologies based on lasers and optical fibres
2. Explain the basic principles of crystal physics
3. Summarise the characteristic and applications of superconducting materials, nanomaterials and smart materials
4. Illustrate the theory of semiconductors and magnetic materials
5. Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

PART A (Answer all questions) (5x2=10)

		Marks	BL	CO	PO
1	a. Briefly explain the term 'Population Inversion'	2	L2	1	1
	b. Discuss any two important applications of Laser	2	L3	1	3
	c. Define 'acceptance angle' and 'Numerical aperture'	2	L2	1	2
	d. Calculate no of atoms/unit cell, radius and Packing density of BCC	2	L3	1	3
	e. what is Bragg's Law	2	L1	1	1

PART B (2x10=20)

		Marks	BL	CO	PO
II	a. With neat diagrams, explain the construction and working of a four level laser system.	7	L2	1	1
	b. Evaluate the ratio of spontaneous emission to stimulated emission at a temperature of 300 K for transition that occur in the visible region (wave length $5500\text{\AA}$) and microwave region (wavelength 1cm)	3	L3	1	2
OR					
III	a. Discuss the structure and propagation through different types of fibre optics	7	L2	1	2
	b. For a step index fibre having core index 1.43 and cladding index 1.4, calculate critical angle, acceptance angle and numerical aperture	3	L3	1	3
IV	a. What is Miller Indices? What are the procedures to find out MI? Draw Miller Indices Planes of (211),(100),(202),(141)	7	L2	2	2
	b. The density of iron is 7.86 g/cc and its atomic weight is 55.85 Calculate the atomic radius, if iron has BCC structure.	3	L3	2	3
OR					
V	a. With neat diagram, explain the working of a Coolidge tube	7	L2	5	1
	b. Electrons are accelerated by 344 volts and are reflected from crystal. The first reflection maximum occurs when glancing angle is 30° , determine the spacing of the crystal	3	L3	5	3